

ATXN2 - Spinocerebellar ataxia type 2 (SCA2)**OMIM:** 183090**OPHRANET:** 98756

MOLECULAR CRITERIA	
GENE	Gene: ATXN2 OMIM: 601517 Gene Description: ATXN2, located on chromosome 12q24.12, encodes ataxin-2, a ubiquitously expressed, predominantly cytoplasmic protein. The key functional domains include a Lsm (Like Sm) domain that mediates RNA binding, a PAM2 (PABP-interacting motif 2) domain interacting with poly(A)-binding protein (PABP), and a large disordered C-terminal region. Ataxin-2 is involved in RNA metabolism, mRNA stability, translation regulation, stress granule assembly/disassembly, and endosomal trafficking. It additionally interacts with TDP-43, a key amyotrophic lateral sclerosis (ALS) pathology protein via stress granule co-localisation, which in cases of ATXN2 dysfunction, can result in increased susceptibility to ALS. Number of Associated OMIM Diseases: 1
DISEASE	Condition [1]: Spinocerebellar ataxia type 2 (SCA2). Alternative Disease Names: Spinocerebellar atrophy II; Olivopontocerebellar atrophy, Holguin Type; Olivopontocerebellar Atrophy II, OPCA2; Spinocerebellar ataxia, cuban type; Cerebellar degeneration with slow eye movements; Wadia-swami syndrome; spinocerebellar degeneration with slow eye movements, SDSEM General Disease Description: Spinocerebellar ataxia type 2 (SCA2) is the second most common autosomal dominant cerebellar ataxia worldwide, representing between 13–15% of all autosomal dominant SCAs. This condition is caused by an abnormal expansion of CAG trinucleotide repeats in exon 1 of the ATXN2 gene. Normal alleles contain ≤ 31 CAG repeats, while complete-penetrance pathogenic alleles are found to have ≥ 35 CAG repeats, with the typical SCA2 range being 36–64 repeats. Pathomechanism: Gain of Function, Expansion Repeats pLI Score: 1 LOEUF Score: 0.32 DECIPHER %HI: 4.16% Intolerant to complete loss of function
INHERITANCE	Autosomal Dominant Genetic anticipation is observed. Repeat length is often expanded in successive generations, particularly in cases of paternal transmission. Increasing repeat length is strongly associated with an earlier onset in children relative to the parent. Penetrance: Variable Incomplete and complete penetrance observed, strongly associated with the number of CAG repeats. Incomplete penetrance is more often observed in individuals with a CAG

repeat number just above the normal range (32-34), while complete penetrance is observed in the population of patients with ≥ 35 repeats.

- ≤ 31 CAG repeats: normal range with no risk of disease
- 32 CAG repeats: penetrance not well established, a small number of healthy controls carry this length. Insufficient evidence is available to classify this expansion as normal or pathogenic.
- 33–34 CAG repeats: incomplete penetrance. Some carriers develop SCA2, parkinsonism without ataxia, or remain unaffected throughout life. This range is the most clinically uncertain. Patients with 33 CAG repeats typically present at age >50 , and are considered “late onset, however some asymptomatic patients with 34 repeats have been reported into later adulthood (Riess O et al., 1997)
- ≥ 35 CAG repeats: Complete penetrance. The primary pathogenic alleles are known to have 37-39 repeats, achieving complete penetrance with longer . Almost all carriers will develop SCA2 if they live long enough (age of onset is dependent on the number of repeats, average is estimated at the fourth decade of life)
- Very large expansions (>64 repeats): Extremely rare; associated with childhood or even infantile onset; one reported case of ~ 884 repeats in a 5-year-old child (Sánchez-Corona et al., 2020)
- Intermediate length repeats (estimated between 30-32) have been recognized as a positive risk factor for the development of ALS.

Expressivity: Variable

Significant variability is observed in the age of onset, the phenotypic spectrum, rate of progression and severity of the patient’s symptoms. It is noted that these factors cannot be predicated by the patient’s family history or through molecular genetic (DNA) testing.

Age of onset: Onset typically occurs within the fourth decade of life, however can range from early childhood (in cases of significant, large-scale expansions) to late adulthood (70s). The largest window of onset occurs in individuals with less than 40 CAG repeats, ranging between 20-60 years of age. In contrast, large repeat sizes (greater than 45) almost always present before age 20. It is estimated that the CAG repeat length explains a certain degree of variability, modifying genetic factors have been suggested to additionally place a role in the phenotypic expression (Hayes et al., 2000)

Phenotypic spectrum within families: Some carriers present with the classic SCA2 phenotype (ataxia, slow saccades, nystagmus and peripheral neuropathy), while other patients present predominantly with dopamine-responsive parkinsonism without prominent cerebellar features, alternatively recognized as the "SCA2 parkinsonian variant".

Severity of specific features: Cognitive impairment, REM sleep behaviour disorder, peripheral neuropathy, and movement disorder features (chorea, dystonia, myoclonus) all vary in prominence between individuals with identical genotypes

	Genetic anticipations: repeat length tends to expand, particularly in paternal transmission, so children of SCA2 fathers often have longer repeats and therefore earlier, more severe disease than their parent
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CLINICAL CRITERIA	
SYSTEMIC INVOLVEMENT	Neurological and Ophthalmic Involvement Primary Affected Tissue: CNS tissue, primarily affecting the cerebellum, brainstem, spinal cord, and cerebral cortex Secondary Affected Tissue: Skeletal muscle, hypotonia and amyotrophy secondary to impaired neurological functioning.
HPO	Core Phenotypic Presentation: progressive cerebellar ataxia, nystagmus, slow saccadic eye movements, with ophthalmoparesis or parkinsonism observed in a subset of patients. Initial symptoms included gait ataxia, typically accompanied by leg cramps. As the condition progresses, individuals often develop a kinetic or postural tremor, decreased muscle tone, decreased tendon reflexes, and abnormal eye movements with slowed saccades progressing to supranuclear ophthalmoplegia. Dystonia, chorea and dementia are additionally common findings in this patient population
COMPLETE SYMPTOM PROFILE	Ocular/bulbar: Dysphagia · Dysmetric saccades · Gaze-evoked nystagmus · Impaired horizontal smooth pursuit · Ophthalmoplegia · Rod-cone dystrophy · Slow saccadic eye movements Motor/tone: Distal amyotrophy · Generalized hypotonia · Hypotonia · Rigidity · Action tremor · Postural tremor · Bradykinesia · Spasticity · Myoclonus · Parkinsonian disorder Cerebellar/coordination: Ataxia · Progressive cerebellar ataxia · Limb ataxia · Dysarthria · Dysdiadochokinesia · Dysmetria · Postural instability · Unsteady gait · Cerebellar atrophy · Olivopontocerebellar atrophy · Spinocerebellar tract degeneration · Dilated fourth ventricle Neurological/other: Babinski sign · Hyporeflexia · Impaired vibratory sensation · Fasciculations · Oculomotor apraxia · Dementia · Urinary incontinence · Urinary bladder sphincter dysfunction
NATURAL HISTORY	Age of Onset: The average age of onset is the fourth decade, with a ten- to 15-year disease duration. Age of onset has been inversely correlated to CAG repeat length, however, cannot explain onset variation in isolation. Progression: SCA2 is classified into 3 distinctive stages, asymptomatic, prodromal, and ataxic. No detectable clinical, paraclinical, and neurophysiological disease features are observed in the asymptomatic phase, with a SARA score of 0. The prodromal stage is

	characterized by initial motor and non-motor abnormalities that do not present definitively as ataxia. This stage appears ~15 years prior to complete ataxia and is accompanied by saccadic slowing and reduced. SARA scores range between 0 and 2 points during this stage. The late ataxic stage is further classified based on the severity of the patient's ataxia (slight, moderate and severe). Slight ataxia presents with early manifestations of the cerebellar syndrome and independent gait, worsening of the prodromal features, RLS/PLMS, and sensory demyelination signs. SARA scores typically between 3 and 14 points. As patients progress to moderate ataxia, they often require permanent support to walk and worsening of non-motor symptoms. SARA scores between 15 and 26 points. Patients with severe ataxia require permanent wheelchair support. SARA scores vary between 27-34, with marked sensory-motor neuronopathy, loss of muscle mass, body fat, and swallowing difficulties. Many patients become bed bound by this stage. SARA scores are typically above 35 points, and patients present with significant sensory conduction blockage, ophthalmoplegia, incontinence, severe insomnia, and mental impairments that often present similarly to dementia (Velázquez-Pérez et al., 2017) Symptom Remission: Not observed Persistent Structural Damage: Post-mortem analyses of patients with SCA2 found an overall reduction in brain size, with marked atrophy of the cerebellum, brainstem, frontal lobe, and cranial nerves. Pallor of the substantia nigra in the midbrain and reduced cerebral and cerebellar white matter volume were additionally observed. Histopathological examination identified significant neuronal loss in the cerebellar Purkinje cell layer, observed with a reduction in dendritic arborization and torpedo-like axonal deformations. Additional findings included loss of inferior olivary neurons in the brainstem, degeneration of pontine and other pre-cerebellar brainstem nuclei, and neuronal reduction in the substantia nigra of the mesencephalon. Widespread neuronal loss was further noted in the cerebral cortex, basal forebrain, thalamus, and spinal cord. Brain imaging of SCA2 patients additionally revealed significant cerebellar volume loss in both gray and white matter, atrophy of the pons, medulla oblongata, spinal cord, parietal cortex, and thalamus (Velázquez-Pérez et al., 2017). Life Limiting: Individuals typically present with symptoms in adulthood and live an estimated 10-20 years following their diagnosis. The primary causes of death are attributed to complications of the disease, including bronchopneumonia, bronchial aspiration, and cardiovascular failures (Velázquez-Pérez et al., 2017). Impact on Longterm Functioning: Due to the progressive nature of the condition, patients decline significantly, reflected by a substantial loss of motor function, muscle control and impaired cognitive functioning. Patients lose their independence, requiring mobility supports during the early phases, followed by complete immobilization by the late stages of the disease. SCA2 poses a significant impact on the patient's overall quality of life, resulting in severe disability and physical limitations.
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THERAPEUTIC WINDOW	WIDE: SCA2 has a long asymptomatic phase, with initial symptoms most often presenting during adulthood. This delay in symptom onset creates a large therapeutic window for treatment intervention, and slow symptom progression or prevent the manifestation of serious complications.
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PRACTICAL CRITERIA	
CURRENT THERAPEUTIC APPROACH	Status: Management remains supportive and directed at symptom management. No targeted therapy has been developed to delay or halt disease progression. The disease demonstrates a more rapid rate of progressive if the onset occurs prior to age 20 years. Response to current treatment is variable. An outline of the current pharmaceutical approaches for symptom management and reduction is outlined below. • Lithium carbonate was assessed in a 12-month double-blind, placebo-controlled trial of 16 patients and produced non-significant changes in SARA scores and brain volume, though it significantly reduced depression scores (BID). • Riluzole, a benzothiazole medication designed to inhibit the release of glutamate, was evaluated in a 12-month trial of 16 patients and was found to significantly decrease SARA scores. • Zinc sulfate was tested in a 6-month double-blind, placebo-controlled trial of 36 patients, and was noted to improve gait, posture, stance, and dysdiadochokinesia, and reduced saccadic latency. • Lisuride, a monoaminergic pharmacotherapy, was assessed in a cohort of 12 patients and was found to decrease the periodic limb movements during sleep (PLMS) index and improve subjective sleep quality. • B vitamins, given at 10,000 U/week for the first four weeks followed by 5,000 U/week through week 12, were evaluated in a 3-month open-label trial of 20 patients and were associated with improvements in sensory neuropathy and painful muscle cramps. Clinical trials have been conducted to evaluate the effectiveness of pharmacotherapies and physiotherapy on symptom progression, however, remain focused on symptom management instead of prevention. Despite the reported trial results highlighted above, follow up studies are recommended to further evaluate the effect of these medications on symptom management. In non-pharmacological clinical trials, cerebellar transcranial direct current stimulation (tDCS) has shown the most promising results (Alberto et al., 2021). A randomized, sham-controlled trial in 61 patients with neurodegenerative ataxia (12 with SCA2) demonstrated that two cycles of cerebellar anodal/spinal cathodal tDCS led to significant improvements in SARA, ICARS, cognition (CCAS scale), and quality of life sustained to 52 weeks. A meta-analysis of 14 RCTs (406 patients) confirmed that cerebellar NIBS

	significantly reduced SARA scores (MD −3.45) and ICARS scores (MD −10.87), with tDCS outperforming rTMS.
CURRENT TARGETED THERAPIES	Status: AVAILABLE - ASO, siRNA, CRISPR ASO: A gapmer antisense oligonucleotide, ASO7, was developed to target aberrant ATXN2 mRNA expression (Scoles et al., 2017). In preclinical studies, a single intracerebroventricular injection was found to reduce ATXN2 levels by 70–80% for more than 10 weeks. Treatment of symptomatic mice improved motor function, normalized Purkinje cell firing frequency, and restored expression of PC-specific proteins (Rgs8, Pcp2, Pcp4, Homer3, Cep76, Fam107b). Efficacy was demonstrated even when treatment was initiated >12 weeks after motor symptom onset, suggesting a wider therapeutic window in established disease. ASO development for SCA2 has not yet progressed to human clinical trials. siRNA: An ongoing clinical trial (NCT06672445) is investigating the effectiveness of ARO-ATXN2, a targeted siRNA for the degradation of ATXN2 mRNA, in adults with SCA2. This is a Phase 1, Placebo-controlled dose escalating study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of ARO-ATXN2. The study is currently active and no longer recruiting participating. A recent study (Amade et al., 2025) demonstrated that an AAV vector with potent motor neuron tropism delivering artificial miRNAs was effective for ATXN2 reduction throughout the CNS (doses 40-fold lower than previously published ASO approaches) in TDP-43 overexpressing ALS mice. A 50% increase in survival, improved strength, reduced motor neuron death, decreased neuroinflammation, and reduced phosphorylated TDP-43 was observed. While this work was conducted in ALS models, the same AAV-RNAi platform may be translated for the treatment of SCA2, given the shared ATXN2 target. The potential for a single administration providing durable gene silencing creates a therapeutic advantage for an AAV-mediated approach, compared to the need for repeated intrathecal ASO injections. CRISPR: A study investigated the use of targeted long-read nanopore sequencing to assess the effectiveness of single- and dual-guide RNA AAV-CRISPR editing of human ATXN2 in transgenic SCA2 mouse models. They reported a 10–25% editing efficiency at the ATXN2 locus, alongside significant unintended outcomes, including AAV integration at the target site (1–2% containing the entire AAV genome), >150 kb deletions, and rearrangements of the transgenic allele. Standard PCR-based sequencing methods failed to detect many of these off-target events, highlighting the need for unbiased long-read sequencing to fully characterize editing outcomes. This is an early, preclinical study that has not progressed to clinical human trials, however, does demonstrate the potential for CRISPR/AAV based therapies. In addition, their findings support the incorporation of comprehensive genomic safety assessments into preclinical development, helping to improve the safety and translational readiness of future CRISPR-based therapies for SCA2.

AVAILABILITY OF PRECLINICAL MODELS	STATUS: AVAILABLE Both transgenic and knockout mouse models of disease have been developed. A comprehensive review of the currently recognized models is outlined by Scoles & Pulst (2018).
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BENEFIT: RISK ASSESSMENT	
DISEASE SEVERITY	SEVERE - SCA2 is a progressive, debilitating condition that results in a reduced life expectancy in late stages of disease. Patients slowly lose their motor control and eventually present with significantly impaired cognitive functioning. Symptoms do not remit and pose a significant impact on the patient's overall quality of life as the disease progresses.
CURRENT THERAPEUTIC ASSESSMENT	LIMITED - treatment for SCA2 remains limited. The clinical gold standard is focused on symptom management, and there does not currently exist a treatment approach that can halt disease progression.
THERAPEUTIC IMPACT ON FUNCTION, QUALITY OF LIFE AND SURVIVAL	SIGNIFICANT - without intervention, SCA2 remains a debilitating condition that eventually results in complete dependence of ambulatory assistance and caregiver support. Without targeted treatment, SCA2 patients are expected to have a shorter life expectancy, with a prognosis of 10-15 years following their initial diagnosis. The development of targeted therapy may prevent progression or development of symptoms and holds the potential to significantly improve the patient's quality of life, independence and longevity.

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